TWO MECHANISMS FOR THE SCHATTEN THRESHOLD FOUR

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> It would be interesting to understand whether the appearance of the same threshold in these two settings reflects some connection between them. In particular, the endpoint case $p = 4$ seems especially interesting, since its Schatten norm can be explicitly computed.

Bampouras, *Failure of Nehari's Theorem for Multiplicative Hankel Forms in Schatten Classes II*, arXiv:2610.06728, end of §1.

Answered: the mechanisms differ. Evidence: Sketch.

The two thresholds arise from different mechanisms. This does not rule out a further relation between the settings.

Proof. Let Γ be either the positive lattice $\mathbb{N}_0^d$ or a bounded open convex set $\Omega \subset \mathbb{R}^n$. In both settings, expanding $\|H_\varphi^* H_\varphi\|_{S_2}^2$ gives the same quartic integral

$$\|H_\varphi\|_{S_4}^4 = \int_{a+d=b+c} \overline{\widehat{\varphi}(a)}\widehat{\varphi}(b)\widehat{\varphi}(c)\overline{\widehat{\varphi}(d)} W_\Gamma(a,b,c,d),$$

where W_Γ is the measure of the allowed common shift, explicitly

$$W_\Gamma(a,b,c,d) = \big|\{\zeta \in \Gamma : a-\zeta,\ b-\zeta,\ \zeta+c-a \in \Gamma\}\big|, \qquad d = b+c-a.$$

Replacing W_Γ by 1 gives $\|\varphi\|_4^4$.

For $\Gamma = \mathbb{N}_0^d$, direct counting gives $W_\Gamma = \prod_{j=1}^d (1 + \min(a_j, b_j, c_j, d_j)) \geq 1$. More precisely, the constant monomial contributes the unweighted quartic form, and each other monomial contributes a nonnegative fourth power. Hence $\|\varphi\|_4 \leq \|H_\varphi\|_{S_4}$. This positive constant-term contribution is the torus mechanism.

For $\Gamma = \Omega$, put $\omega = \chi_\Omega * \chi_\Omega$ and $L_x = \{\xi \in \Omega : x - \xi \in \Omega\}$. If Z is the admissible-shift set, then $Z \subset L_a \cap L_b$. Translation by $c-a$ maps Z into $L_c \cap L_d$, because for $\zeta \in Z$ the point $\zeta' = \zeta + c - a$ satisfies $c - \zeta' = a - \zeta$ and $d - \zeta' = b - \zeta$. Thus its measure is bounded by each of $\omega(a), \omega(b), \omega(c), \omega(d)$. Choose a fixed nonzero $\beta \in C_c^\infty(B(0,1))$ and $\widehat{\varphi}(\xi) = \beta((\xi - z)/\rho)$, with $B(z,\rho) \subset 2\Omega$ and z at distance ρ from the boundary. The Schur test and the Hilbert–Schmidt identity give

$$\|H_\varphi\|_{S_4}^4 \leq a\rho^{3n}\|\beta\|_1^2\|\beta\|_2^2, \qquad \|\varphi\|_4^4 = \rho^{3n}\|\check{\beta}\|_4^4,$$

where $a = \sup_{B(z,\rho)} \omega$. Since ω vanishes on the boundary, $a \to 0$ as $\rho \downarrow 0$. Therefore no uniform $L^4 \leq CS_4$ inequality holds for this Paley–Wiener model.

The Paley–Wiener statement is method-level. Let $\Omega \subset \mathbb{R}^2$ be a bounded convex set with C^2 boundary and positive curvature, with uniform interior and exterior rolling-ball radii. Fix $\beta \in C_c^\infty(B(0,1))$, choose caps $\widehat{\varphi}_i(\xi) = \beta((\xi - z_i)/\rho)$ contained in 2Ω, and require the input lenses

$$D_i = \{\xi \in \Omega : \xi + \eta \in \operatorname{supp} \widehat{\varphi}_i \text{ for some } \eta \in \Omega\}$$

to be pairwise disjoint. For $\psi = \sum_{i=1}^N \varphi_i$ the narrow counting test is $h = \varphi = \psi$, and

$$Q_p(\psi,\psi) = \frac{\|\psi\|_2^2}{\|\psi\|_1 \|H_\psi\|_{S_p}}.$$

Under these hypotheses, $Q_p \leq C_p$ for $1 \leq p \leq 4$, while suitable such configurations make Q_p unbounded for every $p > 4$.

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Here are the estimates giving the planar upper bound. Choose w_0 with $\beta(w_0) \neq 0$, put $x_i = z_i + \rho w_0$, $m_i = x_i/2$, $h_i = \operatorname{dist}(m_i, \partial\Omega)$, and $L_i = \Omega \cap (x_i - \Omega)$. The rolling-ball condition and containment of each cap give $h_i \geq c\rho$; the lenses L_i are disjoint and their areas $\omega_i = |L_i|$ sum to S. Subdivide a fixed central contraction of each lens into disjoint squares of side $s\rho$, with $s > 0$ sufficiently small. There are at least $c\omega_i\rho^{-2}$ such squares. The normalized indicators of a square and its reflection through $x_i/2$ form orthonormal families. On each pair the matrix coefficient of H_ψ has size at least $c\rho^2$: only cap i contributes, and β varies by at most half its nonzero value. The Schatten diagonal inequality therefore yields, for every $p \geq 1$,

$$\|H_\psi\|_{S_p}^p \geq c\rho^{2(p-1)} S. \tag{1}$$

Group the caps by depth: I_j consists of those with $h_i \in [\delta_j, 2\delta_j), \delta_j = 2^j c\rho$, and h_i below a fixed small rolling-ball radius; the remaining deep class has bounded cardinality. For a shallow lens, rolling balls give $|L_i| \geq c\delta_j^{3/2}$ and $L_i \subset \{u : \operatorname{dist}(u, \partial\Omega) \leq 4\delta_j\}$. The area of this boundary layer is $O(\delta_j)$, so disjointness gives $N_j\delta_j^{1/2} \leq C$, where $N_j = |I_j|$. Also $S \geq c\sum_j N_j\delta_j^{3/2}$ and each lens has area at least $c\rho^{3/2}$. Set $A = 1 + \sum_j \sqrt{N_j}$. These bounds imply

$$A^2 \geq cN, \qquad A^8\rho^2 \leq CN^2 S. \tag{2}$$

For the second inequality, write

$$\sqrt{N_j} = N_j^{1/4}(N_j\delta_j^{3/2})^{1/8}(N_j\delta_j^{1/2})^{1/8}\delta_j^{-1/4}$$

and apply Hölder with exponents $4, 8, 8/5$; the geometric sum $\sum_j \delta_j^{-2/5} \leq C\rho^{-2/5}$ gives $\sum_j \sqrt{N_j} \leq CN^{1/4}S^{1/8}\rho^{-1/4}$. The deep class and the additive 1 in A are absorbed using its bounded size and $S \geq c\rho^{3/2}$.

The needed lower Khintchine estimate is

$$\|\psi\|_1 \geq cN^{3/2}A^{-2}. \tag{3}$$

To see why curvature matters, first fix one depth class. The identity $\|\psi_j\|_4^4 = \|\widehat{\psi}_j * \widehat{\psi}_j\|_2^2$ bounds this fourth moment by $C\rho^6 E_j$, where E_j counts quadruples satisfying $|m_i + m_{i'} - m_k - m_{k'}| < 2\rho$. On each of finitely many boundary arcs write the boundary as $(a, g(a))$, with $g'' \geq \kappa > 0$. Disjoint lenses and the boundary-layer area estimate show that an interval of length $\lambda \geq \sqrt{\delta_j}$ contains at most $C\lambda/\sqrt{\delta_j}$ boundary coordinates. For fixed i, k, the quadruple relation puts $a_{k'}$ within $O(\rho)$ of $a_{i'} + a_i - a_k$; its vertical component forces

$$|g(a_i) + g(a_{i'}) - g(a_k) - g(a_{i'} + a_i - a_k)| \leq C\delta_j.$$

Split each depth class into finitely many such arcs and apply the triangle inequality in L^4, so it suffices to count quadruples lying in one graph chart. The derivative of the expression inside absolute values has constant sign and magnitude at least $\kappa|a_i - a_k|$. If $|a_i - a_k| \geq \sqrt{\delta_j}$, the local packing bound leaves only $O(1)$ choices for i', k'. If it is smaller, there are only $O(1)$ choices of k for each i; for each such pair there are at most N_j choices of i', and cap separation leaves only $O(1)$ choices of k'. Thus $E_j \leq CN_j^2$; classes at fixed positive depth contain only $O(1)$ caps and satisfy the same estimate trivially. Consequently $\|\psi_j\|_4 \leq C\sqrt{N_j}\rho^{3/2}$ and, summing classes, $\|\psi\|_4 \leq CA\rho^{3/2}$. Disjoint cap supports give $\|\psi\|_2^2 = N\rho^2\|\beta\|_2^2$. Interpolation $\|\psi\|_2^2 \leq \|\psi\|_1^{2/3}\|\psi\|_4^{4/3}$ proves (3).

Since $\|\psi\|_2^2 \asymp N\rho^2$, (1) and (3) give

$$Q_p(\psi, \psi)^p \leq C_p \frac{A^{2p}\rho^2}{N^{p/2}S}.$$

For $p \leq 4$, (2) bounds the right side by $C_p N^{p/2-2} \leq C_p$. This proves boundedness through the endpoint for this counting family; it does not prove the Paley–Wiener Nehari inequality.

For the reverse threshold, take $\rho = \delta$ caps at depth comparable to δ along a curved boundary arc, with centers separated tangentially by a sufficiently large multiple of $\sqrt{\delta}$. Then $N \asymp \delta^{-1/2}$, the corresponding lenses are disjoint, and $\omega = |\Omega \cap (x - \Omega)| \le C\delta^{3/2}$ on every cap. Apply weighted Cauchy–Schwarz with an integer $k > 1$:

$$\|\psi\|_1 \le \left(\int_{\mathbb{R}^2} (1+\rho^2|x|^2)^{-k}\,\mathrm{d}x\right)^{1/2} \left(\int_{\mathbb{R}^2} (1+\rho^2|x|^2)^{k}|\psi(x)|^2\,\mathrm{d}x\right)^{1/2} \le C\sqrt{N}.$$

The first factor is $O(\rho^{-1})$. By Plancherel, the square of the second factor is bounded by a constant times $\sum_{|\alpha|\le k} \rho^{2|\alpha|}\|\partial^\alpha \widehat{\psi}\|_2^2$. The differentiated cap supports remain disjoint, and each cap contributes $O(\rho^2)$ to this sum; hence it is $O(N\rho^2)$. For one cap the Schur and Hilbert–Schmidt bounds are $\|H_{\varphi_i}\| \le C\rho^2$ and $\|H_{\varphi_i}\|_{S_2}^2 \le C\omega\rho^2$. The operators act on pairwise orthogonal D_i blocks, so for $p \ge 2$,

$$\|H_\psi\|_{S_p}^p = \sum_i \|H_{\varphi_i}\|_{S_p}^p \le CN\delta^{3/2}\rho^{2(p-1)}.$$

Together with $\|\psi\|_2^2 \asymp N\rho^2$, this gives $Q_p \ge cN^{1/2-1/p}(\rho^2/\delta^{3/2})^{1/p} \asymp N^{1/2-2/p}$, which diverges exactly for $p > 4$. The lattice threshold has a different explanation: its positive constant-term contribution to the quartic form gives $\|\varphi\|_4 \le \|H_\varphi\|_{S_4}$, whereas the Paley–Wiener cap construction above rules out the analogous L^4 bound. □

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